

VITISH 2025

(SIH Internal Hackathon)

- **PS ID: SIH25047**
- **PS Title: Disaster Response Drone for Remote Areas**
- **Theme: Robotics and Drones**
- **PS Category: Hardware**
- **Team ID: VIT-865**



PROJECT NIMBUS

Proposed Solution

- *Hybrid VTOL medical drone*
- Vertical take-off + plane-like speed
- AI navigation & obstacle avoidance

Solves problem by

- Works in remote areas with little or no access to Wi-Fi by using LTE IP
- Can take-off in tight spaces and fly long range distances

Innovation & Uniqueness

- Modular Payload Bay
- Self-built avionics (Pixhawk-class)



**Nimbus 007
(Concept)**

TECHNICAL APPROACH

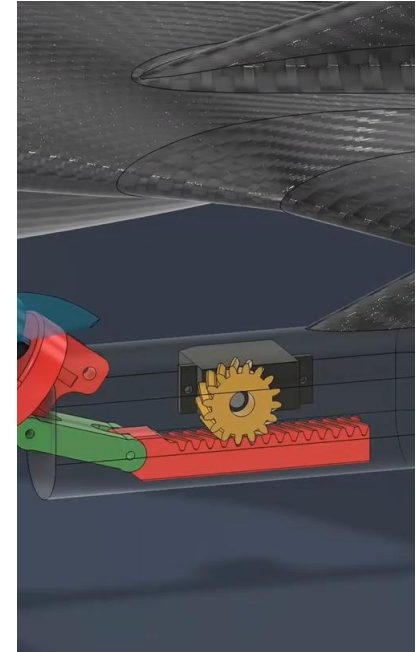
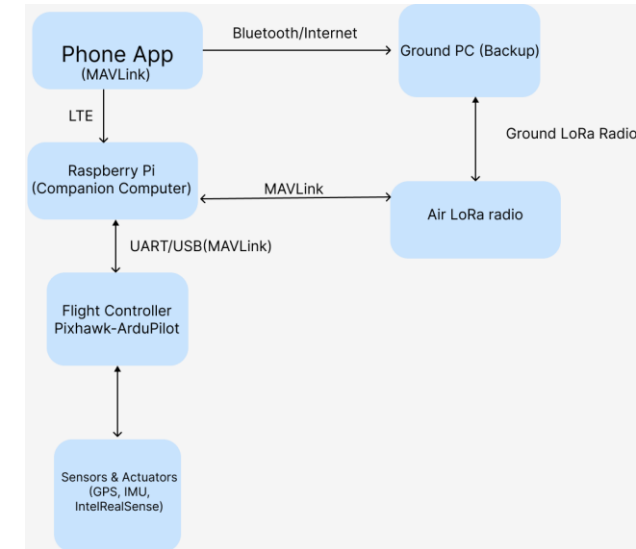
Hardware:

<https://docs.google.com/spreadsheets/d/1F6OGOesjAs7q4XyuNf6iJS6DM2Ykt9MaWld0UDiYUrY/edit?usp=sharing>

- Flight Computer: STM32H743VIH6
- Companion Computer: RaspberryPi
- Tilting rotor mechanism
- Sensors: GPS, LIDAR/Ultrasonic, IntelRealSense Camera
- Medical payload module (compact container)

Software / AI:

- Programming: **Python, C++, React Native for App**
- Flight control: **Pixhawk 6c**
- Simulation: **Gazebo, MATLAB/Simulink**
- AI navigation: **OpenCV, Distance estimation is done by combining YOLOv8 with depth data from IntelRealSense to get accurate 3D positions of detected objects**



FEASIBILITY AND VIABILITY

Feasibility Analysis:

- Technical: 7(+0.5)Kg possible with 4 motors
- Economic: Prototype Cost ~₹1.5 lakh, scalable and cost-efficient in bulk
- Operational: Quick deployment from relief trucks/camps
- Regulatory: Govt. support ensures compliance



Potential Challenges



Battery Limitations → Limited flight range.



Weather Constraints → Strong winds/rain may affect operations.



Navigation Risks → GPS jamming, obstacle-dense areas.



Payload Constraints → Weight and secure handling of medical kits.

Mitigation Strategy



Swappable high-capacity batteries / solar charging stations



Weather-resilient design + fallback emergency landing protocols



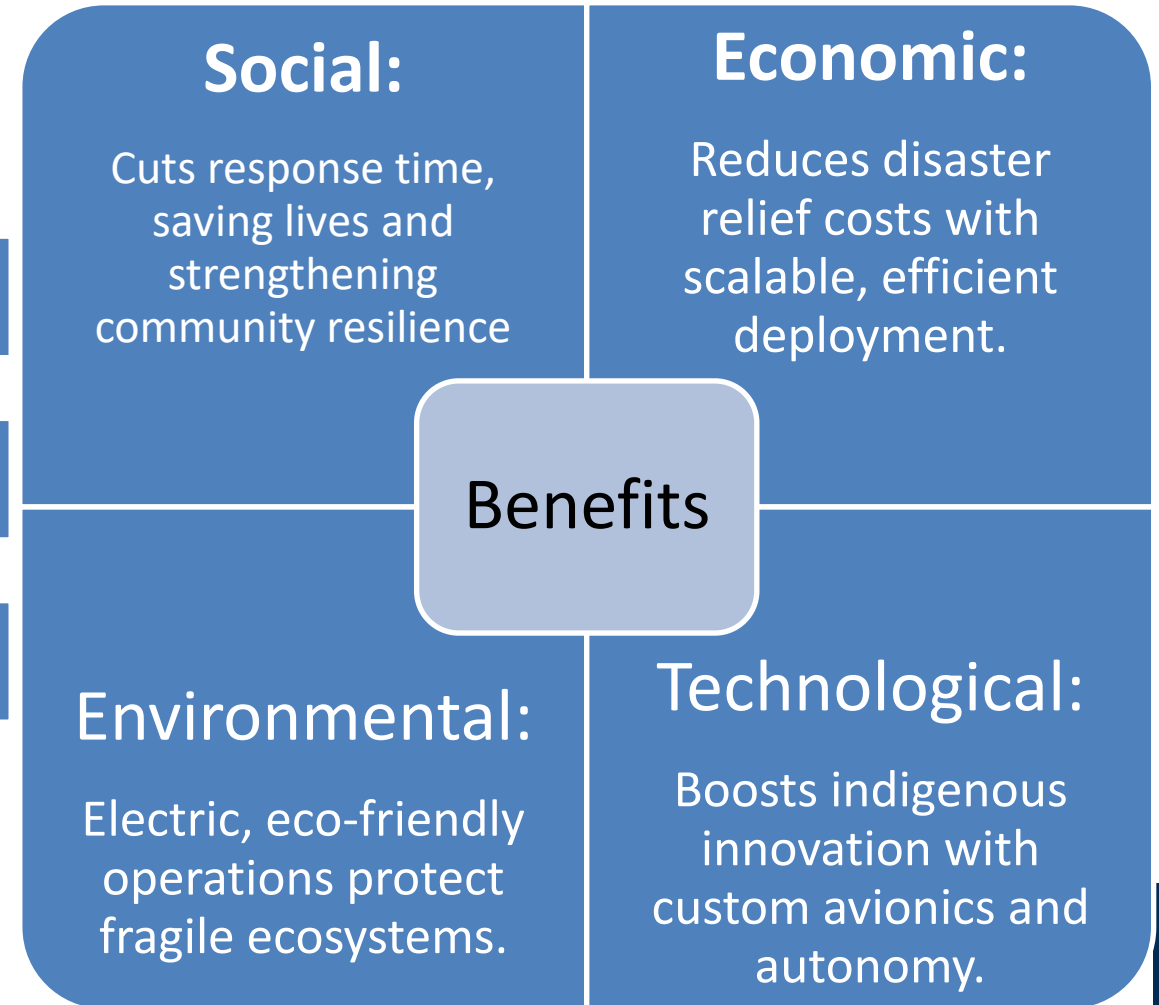
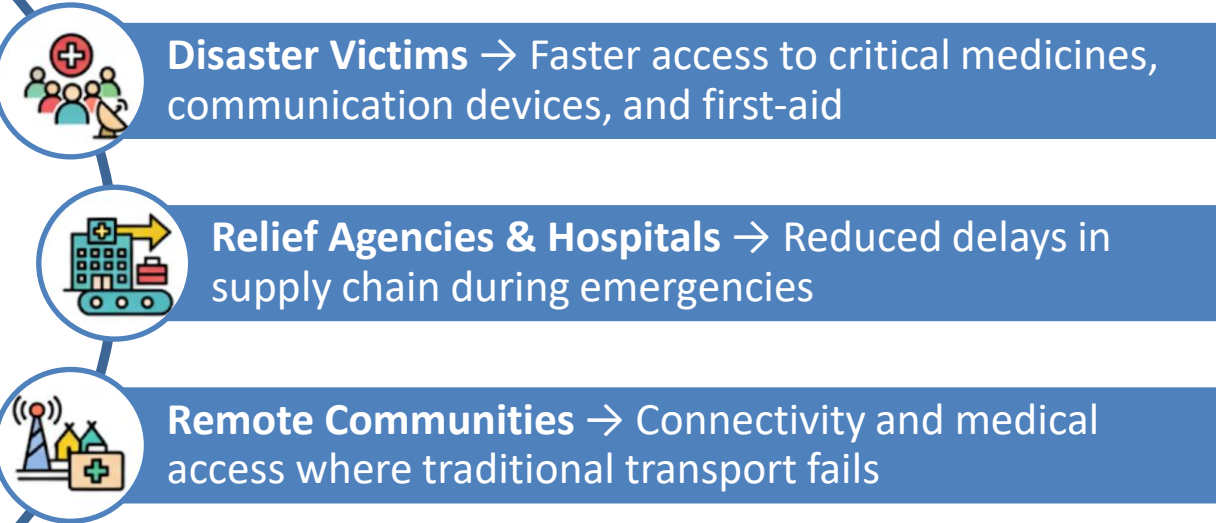
Hybrid navigation (GPS + AI vision + IMU fusion).



Lightweight modular payload containers with shock absorption



Potential Impact on Target Audience



RESEARCH AND REFERENCES

Key References Used

•Disaster Relief Drones – Case Studies

- *UNICEF & WHO reports on UAVs for medical supply delivery.*

•Hybrid VTOL Drone Research

- *IEEE Xplore papers on VTOL efficiency & hybrid propulsion.*

•Navigation & AI in Drones

- *MIT research on AI-based obstacle avoidance.*

•Power & Control Systems

- *Electronics tutorials (Arduino, sensors, motor drivers) + LTSpice simulations.*

Links / Resources

- [WHO – Drones for Health Delivery](#)
- <https://www.zipline.com/>
- IEEE Xplore – UAV & VTOL Papers
- [NASA – UAV Navigation Research](#)
- KiCAD Circuits & LTSpice Simulations